

RadPi Model Comparison

RadPi Project
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RadPi Model Comparison

Edge-Deployable Medical Imaging Models for Raspberry Pi 5

Comprehensive cross-modality survey of pretrained, INT8-quantized models that fit a Raspberry Pi 5 inference target across six clinical domains: brain MRI, chest X-ray, bone fracture X-ray, dental panoramic X-ray, dermatology, and fundus/retinal imaging.

Hardware Target

Spec	Value
Board	Raspberry Pi 5
CPU	Broadcom BCM2712, quad-core ARM Cortex-A76 @ 2.4 GHz
RAM	8 GB LPDDR4X-4267
Architecture	ARM64 (aarch64)
Accelerator	None (CPU-only inference)
OS	Raspberry Pi OS 64-bit (Debian Bookworm)
Runtime	ONNX Runtime 1.17+ (ARM64 build) or TFLite Runtime 2.15+
Quantization	INT8 post-training quantization mandatory for all models
Performance envelope	100-300 ms per inference for MobileNet/EfficientNet-B0/YOLOv8n class models at 224x224; 0.5-2 s per image for compact U-Net segmentation at 128x256

Every model listed below is selected against this envelope. 3D networks (e.g. MONAI SegResNet, BraTS 3D U-Net) are explicitly excluded for memory reasons. Vision transformers larger than ViT-Tiny / SigLIP-base are excluded for latency reasons.

Deployment Pipeline

The conversion path is identical across modalities and is the single most important standardization decision in the project.

PyTorch path:

```
.pt or .pth -> torch.onnx.export(model, dummy, "model.onnx", opset_version=17)
            -> onnxruntime.quantization.quantize_dynamic(per_channel=True, weight_type=QInt8)
            -> ort.InferenceSession("model.int8.onnx", providers=["CPUExecutionProvider"])
```

Keras / TensorFlow path:

```
.h5 or .keras -> tf.lite.TFLiteConverter.from_keras_model(model)
            -> converter.optimizations = [tf.lite.Optimize.DEFAULT]
            -> converter.representative_dataset = rep_data_gen # for full INT8
            -> tflite_runtime.interpreter.Interpreter("model.int8.tflite")
```

Ultralytics YOLO path:

```
model.export(format="onnx", imgsz=640, opset=17, simplify=True, int8=True)
```

Expected speedups from INT8: 2-4x size reduction, 2-3x latency improvement on Cortex-A76 vs FP32.

Executive Summary

Primary pick per modality. Full per-modality tables follow.

RadPi Model Comparison

Modality	Task	Best model	Arch	Size	Format	License	Est. inference
Brain MRI	4-class tumor	alanjafari/BrainTumorAI	YOLOv8n	14 MB	.pt -> ONNX	MIT	<1 s
Brain MRI	Tumor segmentation	wdika/SEG_UNet_BraTS2023AdultGlioma	AttentionUNet	~30 MB	.atommic -> PyTorch	Apache-2.0	1-2 s
Chest X-ray	18-pathology classification	mlmed/torchxrayvision densenet121-res224-all	DenseNet121 (7.9M)	~30 MB	.pt	Apache-2.0	1-2 s
Chest X-ray	Lung field segmentation	imlab-iiip/lung-segmentation-2d	U-Net	~30 MB	.hdf5 (Keras)	none stated	1-2 s
Bone fracture	Detection	RuiyangJu/Bone_Fracture_Detection_YOLOv8	YOLOv8n	~12 MB	.pt -> ONNX	AGPL-3.0	<1 s (3-10 FPS)
Bone fracture	Segmentation	None publicly available	--	--	--	--	use Grad-CAM
Dental	Tooth segmentation	SerdarHelli Teeth U-Net	Keras U-Net	161 MB	.h5 -> TFLite	none stated	1-3 s
Dental	Pathology detection	SubGlitch1/DentalXrayAI	YOLOv8n	~6 MB	.pt -> ONNX	none stated	<1 s
Dermatology	Lesion classification	HugsVision/Skin-Cancer MobileNetV2	MobileNetV2	9.2 MB	.pth -> ONNX	Apache-2.0	~150 ms
Dermatology	Lesion segmentation	Ahmed-Selem/Shifaa-Skin-Cancer-UNet	Tiny U-Net (128px)	<5 MB	PyTorch	MIT	<1 s
Fundus	DR 5-class grading	EfficientNet-B0 on APTOS (train from notebooks)	EfficientNet-B0	~20 MB	.pt -> TFLite	varies (MIT/Apache)	0.5-1 s
Fundus	Retinal vessel segmentation	orobix/retina-unet	Patch U-Net (48x48)	<10 MB	Keras	none stated	<1 s

1. Brain MRI

Clinical task: 4-class tumor classification (glioma, meningioma, pituitary, no tumor) on T1-CE axial slices, plus per-pixel tumor segmentation. Input is a single 2D slice. 3D volumetric BraTS pipelines (SegResNet, nnU-Net 3D) are out of scope for the Pi 5.

1A. CLASSIFICATION

Model	URL	Arch	Params / Size	Format	License	Dataset
alanjafari/BrainTumorAI	huggingface.co/alanjafari/BrainTumorAI	YOLOv8n classifier	3.2M / 14 MB	.pt -> ONNX	MIT	Msoud Nickparvar (7,023 imgs)
ShabGaming/Brain_MRI_Tumor_Classification	huggingface.co/ShabGaming/Brain_MRI_Tumor_Classification	CNN	~25 MB	.pt	varies	Kaggle MRI
hassaanik/Tumor_Detection	huggingface.co/hassaanik/Tumor_Detection	CNN	small	.pt	varies	binary tumor / no-tumor
MobileNetV2 fine-tune notebook (Kaggle, slashblade)	kaggle.com/code/slashblade/brain-tumor-classification-mobilenet-v2	MobileNetV2 transfer	3.5M / ~14 MB	.h5 -> TFLite	notebook (Apache-style)	Msoud Nickparvar
google/vit-base-patch16-224 fine-tunes (various HF forks)	huggingface.co/models?search=brain+mri+vit	VIT-Base	86M / ~340 MB	.pt	Apache-2.0	varies

Recommendation. Ship `alanjafari/BrainTumorAI` as the default. 14 MB, MIT-licensed, exports cleanly to ONNX, gives sub-second inference. Keep the Kaggle MobileNetV2 notebook in reserve for cases where you want full INT8 TFLite from the ground up or need to retrain on local data.

1B. SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi fit
wdika/SEG_UNet_BraTS2023AdultGlioma	huggingface.co/wdika/SEG_UNet_BraTS2023AdultGlioma	2D AttentionUNet	~30 MB	.atom-mic -> PyTorch	Apache-2.0	BraTS 2023 Adult Glioma	Dice ~0.88 (whole tumor)	Of (w ing 2 :
Mehrdad-Noori/Brain-Tumor-Segmentation	github.com/Mehrdad-Noori/Brain-Tumor-Segmentation	2D U-Net (Keras)	~10M	training code only	MIT	LGG MRI Segmentation	Dice 0.91	Of (e: ce ler
adityajn105/brain-tumor-segmentation-unet	github.com/adityajn105/brain-tumor-segmentation-unet	U-Net (Keras)	~8M	training code only	MIT	LGG FLAIR	Dice 0.90	Of (g
sdsuhajitdas/Brain-Tumor-Segmentation	github.com/sdsuhajitdas/Brain-Tumor-Segmentation	U-Net (PyTorch)	~10M	training code only	MIT	LGG	Dice ~0.89	Of (g
MONAI SegResNet (BraTS)	github.com/Project-MONAI/MONAI	3D SegResNet	~50M	.pt	Apache-2.0	BraTS 3D volumes	Dice 0.92 (WT)	NC (3l >8 pe

Recommendation. Use `wdika/SEG_UNet_BraTS2023AdultGlioma` as the only segmentation model with downloadable checkpoints in this list. Convert to TorchScript at packaging time so deployments don't carry the ATOMMIC dependency. If license clarity matters more than ready weights, train the `adityajn105` U-Net on LGG FLAIR (a half-day on a single GPU) and ship a TFLite INT8 build.

DATASETS

Dataset	URL	Size
Brain Tumor MRI Dataset (Msoud Nickparvar)	kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset	7,023 imgs, 4 classes
Figshare Brain Tumor (Jun Cheng) + masks	figshare.com/articles/dataset/brain_tumor_dataset/1512427	3,064 T1-CE + masks
LGG MRI Segmentation (Mateusz Buda)	kaggle.com/datasets/mateuszbudalgg-mri-segmentation	3,929 FLAIR + masks
BraTS 2020 Training Data	kaggle.com/datasets/awsaf49/brats2020-training-data	371 multimodal 3D
Simezu/brain-tumour-MRI-scan (HF mirror)	huggingface.co/datasets/Simezu/brain-tumour-MRI-scan	7,023 imgs

2. Chest X-ray

Clinical task: multi-pathology classification, binary pneumonia screening, lung field segmentation, and tuberculosis detection on AP/PA chest radiographs.

2A. CLASSIFICATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Report metric
mlmed/torchxrayvision densenet121-res224-all	github.com/mlmed/torchxrayvision	DenseNet121	7.9M / ~30 MB	.pt	Apache-2.0	NIH + CheXpert + RSNA + MIMIC combined	AUC 0.78-0. across pathologies
ayushirathour/chest-xray-pneumonia-detection	huggingface.co/ayushirathour/chest-xray-pneumonia-detection	MobileNetV2	3.5M / ~14 MB	.h5	MIT	Mooney Pneumonia	96% accuracy
keremberke/yolov8m-chest-xray-classification	huggingface.co/keremberke/yolov8m-chest-xray-classification	YOLOv8m (cls)	~50 MB	.pt	AGPL-3.0	binary normal/pneumonia	~95%
ishans24/covid19-detection-xray	huggingface.co/ishans24/covid19-detection-xray	VGG19	~80 MB	.h5	varies	COVID/Normal/Pneumonia 3-class	~93%
MMClassification CheXNet reference	github.com/arnoweng/CheXNet	DenseNet121	7M / ~28 MB	.pth	MIT	NIH ChestX-ray14	AUC 0. (14 pat)

Recommendation. `torchxrayvision densenet121-res224-all` is the single most clinically comprehensive option for Pi deployment. Use it as the default. Layer `ayushirathour MobileNetV2` as a faster front-line pneumonia screen if the device is operating in low-power mode and only needs a binary readout.

2B. LUNG SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	Go
ianpan/chest-x-ray-basic	huggingface.co/ianpan/chest-x-ray-basic	EffNetV2-S + U-Net	22.2M	safetensors	none stated	CheXpert + NIH + CheXmask	Dice 0.943-0.957 (lungs+heart)	OK (good, 1-2 s)	Seq right left anc no cer
imlab-uiip/lung-segmentation-2d	github.com/imlab-uiip/lung-segmentation-2d	Standard U-Net	~10M / ~30 MB	.hdf5 (Keras)	none stated	Montgomery + JSRT	Dice ~0.97	OK (excellent)	Pre tr moc f5 the trai nee
IlliaOvcharenko/lung-segmentation	github.com/IliaOvcharenko/lung-segmentation	U-Net + VGG11 encoder	~25M	.pt	none stated	Montgomery + Shenzhen	Dice 0.961	OK (good)	Py1 cor ON

Recommendation. Prefer `imlab-uiip/lung-segmentation-2d` for true plug-and-play deployment. The repo ships a pretrained `trained_model.hdf5`, convert it directly to TFLite INT8. If you need lungs *and* heart segmented (cardiothoracic ratio computation), switch to `ianpan/chest-x-ray-basic` and accept the missing-license risk.

2C. TB CLASSIFICATION

No standalone pretrained TB model with downloadable weights was located on HuggingFace, Kaggle, or GitHub releases. The recommended approach is to fine-tune MobileNetV2 or EfficientNet-B0 on the combined Montgomery + Shenzhen datasets (~800 images). `torchxrayvision` indirectly detects TB-related radiographic patterns (infiltration, consolidation) without dedicated TB labels.

DATASETS

Dataset	URL	Size
Chest X-Ray Images (Pneumonia) - Paul Mooney	kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia	5,863 imgs
NIH ChestX-ray14	kaggle.com/datasets/nih-chest-xrays/data	112,120 imgs, 14 labels
CheXpert (Stanford)	aimi.stanford.edu/datasets/chexpert-chest-x-rays	224,316 CXRs
COVID-19 Radiography Database	kaggle.com/datasets/tawsifurrahman/covid19-radiography-database	~21K imgs
Montgomery County TB CXR	kaggle.com/datasets/raddar/tuberculosis-chest-xrays-montgomery	138 CXRs + masks
Shenzhen Hospital TB CXR	lhncbc.nlm.nih.gov/...Shenzhen-Hospital-CXR-Set	662 CXRs
TB Chest X-ray (Tawsifur Rahman)	kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset	4,200 imgs

3. Bone Fracture X-ray

Clinical task: fracture detection on extremity (wrist, hand, ankle, etc.) radiographs. Localization via bounding boxes is the standard output; per-pixel segmentation has no public weights.

3A. DETECTION / CLASSIFICATION

Model	URL	Arch	Params / Size	Format	License	Dataset
RuiyangJu/Bone_Fracture_Detection_YOLOv8	github.com/RuiyangJu/Bone_Fracture_Detection_YOLOv8	YOLOv8n	3.2M / ~12 MB	.pt	AGPL-3.0	GRAZPEE DX (20,32 wrist X-ray)
nandodeomkar/autotrain-bone-fracture-detection	huggingface.co/nandodeomkar/autotrain-bone-fracture-detection	Swin-Tiny	~28M / ~110 MB	.pt	Apache-2.0	classification only
prithivMLmods/Bone-Fracture-Detection	huggingface.co/prithivMLmods/Bone-Fracture-Detection	SigLIP2-base (Transformer)	~94M FP32 / ~95 MB INT8	safetensors	Apache-2.0	mixed
mo26-web/Bone-Fracture-Classification	github.com/mo26-web/Bone-Fracture-Classification	MobileNetV3	5.4M / ~22 MB	training code only	varies	mixed
mdciri/YOLOv7-Bone-Fracture-Detection	github.com/mdciri/YOLOv7-Bone-Fracture-Detection	YOLOv7-p6	~80M	.pt + ONNX	GPL-3.0	wrist + har

Recommendation. `RuiyangJu/Bone_Fracture_Detection_YOLOv8` is the standout: ready weights, smallest footprint, real bounding-box output. The AGPL license is the only flag - if RadPi will be distributed as a closed-source product, train the `mo26-web` MobileNetV3 on the same GRAZPEDWRI-DX dataset and ship under a permissive license.

3B. SEGMENTATION: EXPLICIT GAP

No publicly available pretrained fracture segmentation model with downloadable weights exists. Confirmed across HuggingFace, Kaggle, GitHub releases, and TensorFlow Hub. The `FracAtlas` GitHub repo contains a YOLOv8s_seg training notebook with COCO-format segmentation masks but no published weights. Two recommended workarounds:

1. **Grad-CAM on the classification model.** Run YOLOv8n + Grad-CAM++ on the detection head to generate localization heatmaps. Minimal extra compute, ships with `pytorch-grad-cam`.
2. **Train YOLOv8n-seg on FracAtlas.** 4,083 images with masks. Half a day on a single GPU. Ship the resulting `.pt` as the project's first original artifact.

DATASETS

Dataset	URL	Size
GRAZPEDWRI-DX	figshare.com/articles/dataset/GRAZPEDWRI-DX/14825193	20,327 wrist X-rays + bboxes
FracAtlas	figshare.com/articles/dataset/FracAtlas/22717560	4,083 imgs (717 fractured) + seg masks
Bone Fracture Multi-Region (Kaggle)	kaggle.com/datasets/bmadushanirodrigo/fracture-multi-region-x-ray-data	10,580 imgs
MURA (Stanford)	stanfordmlgroup.github.io/competitions/mura	40,561 imgs, 7 upper-extremity types
FracAtlas on HuggingFace	huggingface.co/datasets/yh0701/FracAtlas_dataset	HF mirror

4. Dental Panoramic X-ray

Clinical task: tooth-by-tooth segmentation on panoramic radiographs and caries / pathology detection. Most underserved of the six modalities; only two production-ready model families exist.

4A. TOOTH SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	Gotcha
SerdarHelli Teeth U-Net	huggingface.co/SerdarHelli/Segmentation-of-Teeth-in-Panoramic-X-ray-Image-Using-U-Net	Keras U-Net	~40M / 161 MB	.h5 -> TFLite	none stated	SerdarHelli panoramic (116 imgs + masks)	Dice ~0.93 (binary teeth)	OK (warning, 1-3 s)	Binary teeth-vs-background only, not per-tooth; 161 MB is large but quantizes to ~40 MB INT8
devichand579/Instance_seg_teeth (BB-UNet)	github.com/devichand579/Instance_seg_teeth	YOLOv8 + U-Net	varies	training code only	varies	OdontoAI	per-tooth instance seg	OK (warning)	Two-stage pipeline; heavier to maintain
lvisionLab/deep-dental-image	github.com/lvisionLab/deep-dental-image	Mask R-CNN (Detectron2)	~50M / 350-500 MB	.pt	Apache-2.0	DNS / OdontoAI	high	NO (too big)	Detectron2 stack is heavy; out of scope

Recommendation. SerdarHelli is the only ready-to-use option. Ship the 161 MB `dental_xray_seg.h5`, convert to TFLite INT8 (~40 MB), accept binary teeth segmentation as the v1 capability. Per-tooth FDI numbering needs a second-stage classifier or the devichand579 two-stage pipeline.

4B. PATHOLOGY DETECTION

Model	URL	Arch	Params / Size	Format	License	Dataset	Repc metri
SubGlitch1/DentalXrayAI	github.com/SubGlitch1/DentalXrayAI	YOLOv8n	3.2M / ~6 MB	.pt -> ONNX	none stated	DENTEX (MICCAI 2023)	mAP: 0.458 pathc class
AndreyGermanov/yolov8_caries_detector	github.com/AndreyGermanov/yolov8_caries_detector	YOLOv8n	3.2M / ~6 MB	.pt -> ONNX	varies	caries-only	mAP: ~0.58
AI-RESEARCHER-2024/AI-in-Dentistry	huggingface.co/AI-RESEARCHER-2024/AI-in-Dentistry	MobileNetV2 cls	3.5M / ~14 MB	.h5	varies	oral diseases	~90% acc

Recommendation. `SubGlitch1/DentalXrayAI` for any pathology localization need - covers four DENTEX classes (caries, periapical lesions, deep caries, impacted teeth). Pair with `AndreyGermanov/yolov8_caries_detector` if caries-specific sensitivity matters. Both are ~6 MB and run in well under a second after ONNX INT8 quantization.

DATASETS

Dataset	URL	Notes
DENTEX (MICCAI 2023)	huggingface.co/datasets/ibrahimhamamci/DENTEX	2,332 panoramic X-rays, hierarchical annotations
OdontoAI Open Panoramic Radiographs	github.com/lvisionLab/OdontoAI-Open-Panoramic-Radiographs	4,000 imgs, instance seg
DNS Panoramic Images	github.com/lvisionLab/dns-panoramic-images	543 imgs, COCO format
Tufts Dental Database	tdd.ece.tufts.edu	1,000 panoramic, multimodal
Teeth Segmentation (Kaggle)	kaggle.com/datasets/humansintheloop/teeth-segmentation-on-dental-x-ray-images	seg masks
Oral Diseases (Kaggle)	kaggle.com/datasets/salmansajid05/oral-diseases/data	classification

5. Dermatology

Clinical task: melanoma / non-melanoma skin lesion classification on dermoscopy or smartphone photos, plus lesion boundary segmentation. Richest model ecosystem of the six modalities.

5A. CLASSIFICATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit
HugsVision/Skin-Cancer MobileNetV2	huggingface.co/spaces/HugsVision/Skin-Cancer	MobileNetV2	2.3M / 9.2 MB	.pth -> ONNX	Apache-2.0	HAM10000	~88% acc	OK (excellent, ~150 ms)
junaid54541/Skin-Cancer-Classification-Tflite-Model	github.com/junaid54541/Skin-Cancer-Classification-Tflite-Model	MobileNet variant	~5 MB	.tflite	varies	HAM10000	~85%	OK (excellent, ~100 ms)
lizardwine/Melanoma-003	huggingface.co/lizardwine/Melanoma-003	MobileNetV2	~14 MB	.keras	Apache-2.0	melanoma binary	~90%	OK (excellent)
syaha/skin_cancer_detection_model	huggingface.co/syaha/skin_cancer_detection_model	CNN (Keras)	~30 MB	.h5	varies	ISIC	~87%	OK (good)
AnwarKh1/Skin_Cancer-Image_Classification	huggingface.co/AnwarKh1/Skin_Cancer-Image_Classification	ViT-base	86M / ~340 MB	.pt	varies	HAM10000	~90%	NO (too big)
hasibzunair/melanet	huggingface.co/hasibzunair/melanet	EfficientNet-B0	~20 MB	.pt	varies	binary	~92%	OK (good)

Recommendation. `HugsVision/Skin-Cancer MobileNetV2` at 9.2 MB is the optimal model. For zero-conversion deployment, `junaid54541` ships a `.tflite` directly - run with `tflite_runtime.interpreter` and skip the conversion step entirely. Both produce 100-300 ms latency on Pi 5.

5B. LESION SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	Gotcha
Ahmed-Selem/Shifaa-Skin-Cancer-UNet	huggingface.co/Ahmed-Selem/Shifaa-Skin-Cancer-UNet	Tiny U-Net (16->32->64->128), 128x128 grayscale	<1M / <5 MB	PyTorch	MIT	ISIC subset	Dice 0.9175	OK (excellent, <1 s)	Lightest segmentation model in this entire survey
JCruan519/MALUNet	github.com/JCruan519/MALUNet	Multi-Attention Lightweight U-Net	~1M	training code only	varies	ISIC	Dice 0.91	OK (excellent)	Purpose built lightweight from BIL 2022; try yourself
DevBhuyan/Skin-Lesion-Segmentation	huggingface.co/DevBhuyan/Skin-Lesion-Segmentation	U-Net + DenseNet encoder	~30 MB	.h5 (Keras)	varies	ISIC	Jaccard 0.9086	OK (good)	Heavier than Shifaa, margins more accurate
martinijfb/Skin-Lesion-Segmentation	github.com/martinijfb/Skin-Lesion-Segmentation	U-Net	~15 MB	.pt	varies	ISIC	Dice ~0.89	OK (good)	Weights live in <code>Model_L</code> folder of the repo
VNOpenAI/skin-lesion-segmentation	github.com/VNOpenAI/skin-lesion-segmentation	U-Net + DoubleUNet	varies	Keras + ONNX export	varies	ISIC	Dice ~0.90	OK (good)	Ships ONNX export script

Recommendation. `Ahmed-Selem/Shifaa-Skin-Cancer-UNet` for maximum speed on Pi - 128x128 grayscale input keeps it well under 1 s. Upgrade to `MALUNet` if you need higher accuracy and are willing to train (1-2 hours on a GPU).

DATASETS

Dataset	URL	Notes
HAM10000 (Kaggle)	kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000	10,015 dermoscopy imgs, 7 classes
HAM10000 (Harvard Dataverse)	dataverse.harvard.edu/...doi:10.7910/DVN/DBW86T	canonical source
ISIC Challenge Data (all years)	challenge.isic-archive.com/data	up to 33K images
PH2 Dataset	fc.up.pt/addi/ph2/database.html	200 dermoscopy imgs
DermNet (Kaggle)	kaggle.com/datasets/shubhamgoel27/dermnet	broad clinical dermatology

6. Fundus / DR / Cataract

Clinical task: diabetic retinopathy 5-class grading, multi-disease ocular classification (cataract, glaucoma, AMD), retinal vessel segmentation, and optic disc/cup segmentation.

6A. DIABETIC RETINOPATHY CLASSIFICATION (5-CLASS)

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	G
jdelgado2002/diabetic_retinopathy_detection	huggingface.co/jdelgado2002/diabetic_retinopathy_detection	fastai ResNet-50	25M / ~98 MB	.pkl	MIT	APTOS 2019	Quadratic Kappa 0.84	OK (warning, big)	fa a P lo 1 t. — n 1 pi ty
APTOS competition solutions (mikelkl)	github.com/mikelkl/APTOS2019	EfficientNet-B5 / B0	varies	training code only	MIT	APTOS 2019	Kappa 0.92 (winner)	OK (excellent if EfficientNet-B0)	R b. M
ArjTheHacker/diabetic-retinopathy-detection	huggingface.co/ArjTheHacker/diabetic-retinopathy-detection	CNN	~30 MB	.pt	varies	APTOS	~85% acc	OK (good)	L ci
khornlund/aptos2019	github.com/khornlund/aptos2019	EfficientNet	training code only	MIT	APTOS	high	OK (excellent if B0)	Solid training pipeline	
tahsin314 APTOS solution	github.com/tahsin314/AP-TOS2019	mixed	training code only	MIT	APTOS	high	OK (good)	Another competition reference	

Recommendation. `jdelgado2002/diabetic_retinopathy_detection` is the only download-ready DR-classification weight, but the 98 MB ResNet-50 + fastai pickle is awkward. Best long-term path: retrain EfficientNet-B0 on APTOS 2019 using the `mikelkl` notebook (~5.3M params, ~20 MB INT8 TFLite, sub-second Pi inference). Avoid the pickle in production.

6B. OCULAR DISEASE CLASSIFICATION (CATARACT, GLAUCOMA, AMD)

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	
JordiCorbilla/ocular-disease-deep-learning	github.com/JordiCorbilla/ocular-disease-intelligent-recognition-deep-learning	InceptionV3 (~92 MB) + VGG	~92 MB each	.h5	varies	ODIR-2019, 8 classes	F1 ~0.75	OK (warning, heavy)	
talhaanwarch/ODIR2019 (EfficientNetB3)	github.com/talhaanwarch/ODIR2019	EfficientNet-B3	~12M / ~48 MB	.h5	varies	ODIR-2019	F1 0.901	OK (excellent)	
dduwa/ocular-disease-recognition	github.com/dduwa/ocular-disease-recognition	MobileNet / EfficientNet / VGG	varies	.h5	varies	ODIR-5K	varies	OK (excellent if MobileNet variant)	

RadPi Model Comparison

Recommendation. `talhaanwarch/ODIR2019` EfficientNet-B3 single-eye at ~48 MB. ODIR-2019 is dual-eye natively but a single-eye fork avoids the dual-input plumbing on-device. Convert to TFLite INT8 -> ~15 MB, sub-second inference.

6C. RETINAL VESSEL SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric
orobix/retina-unet	github.com/orobix/retina-unet	Patch-based 48x48 U-Net	~500K / <5 MB	Keras	varies	DRIVE	AUC 0.979
VesselSeg-Pytorch (lee-zq)	github.com/lee-zq/VesselSeg-Pytorch	UNet / Dense U-Net / LadderNet	varies, ~5-15 MB	.pt	varies	DRIVE + STARE + CHASE_DB1	Sens 0.81+
TommyGiak/retinal_vessel_segmentation	github.com/TommyGiak/retinal_vessel_segmentation	U-Net	small	.pt	varies	DRIVE	Dice ~0.81
Retina-Seg ensemble (vineet1992)	github.com/vineet1992/Retina-Seg	ensemble of U-Nets	~50 MB	.pt	varies	DRIVE	Dice ~0.82

Recommendation. `orobix/retina-unet` for absolute minimum footprint. If you need cross-dataset robustness (DRIVE + STARE + CHASE_DB1), use `VesselSeg-Pytorch` and ship the LadderNet checkpoint.

6D. OPTIC DISC / CUP SEGMENTATION

Model	URL	Arch	Params / Size	Format	License	Dataset	Reported metric	Pi 5 fit	Gotcha
seva100/optic-nerve-cnn	github.com/seva100/optic-nerve-cnn	Keras U-Net	small	.h5 in <code>models_weights/</code>	varies	RIM-ONE / Drishti-GS	Dice 0.93+	OK (good)	Plug-and-play
HxFu/AGNet	github.com/HxFu/AGNet	Attention-Gated U-Net (PyTorch)	~10M	.pt	varies	REFUGE / Drishti-GS	Dice 0.94	OK (good)	Convert to ONNX

Recommendation. `seva100/optic-nerve-cnn` for Keras shops, `HxFu/AGNet` for PyTorch shops. Both are small enough to ship alongside the vessel-segmentation model without crowding the 8 GB budget.

DATASETS

Dataset	URL	Notes
APTOS 2019 Blindness Detection	kaggle.com/competitions/aptos2019-blindness-detection	5-class DR
Kaggle DR Detection (EyePACS 2015)	kaggle.com/competitions/diabetic-retinopathy-detection	88K imgs
ODIR-5K	kaggle.com/datasets/andrewmvd/ocular-disease-recognition-odir5k	8-class ocular disease
DRIVE	drive.grand-challenge.org	vessel seg, 40 imgs
STARE	cecas.clemson.edu/~ahoover/stare	vessel seg, 20 imgs
CHASE_DB1	blogs.kingston.ac.uk/retinal/chasedb1	vessel seg, 28 imgs
IDRiD	kaggle.com/datasets/mariaherrero/idrid-dataset	DR + lesion seg
Eye Disease Dataset (HF)	huggingface.co/datasets/Falah/eye-disease-dataset	mixed

Gaps and Cautions

These are the items that will trip up an integration team and should be tracked as project risks.

- Bone fracture segmentation has no public pretrained weights.** Confirmed across HuggingFace, Kaggle, GitHub releases, and TensorFlow Hub. The only path forward is to train YOLOv8n-seg on FracAtlas (4,083 images with masks) or ship Grad-CAM heatmaps from the YOLOv8n detection model as a localization proxy.
- Tuberculosis classification has no standalone pretrained model.** TorchXRayVision detects TB-related patterns (infiltration, consolidation) indirectly. For dedicated TB inference, fine-tune MobileNetV2 or EfficientNet-B0 on combined Montgomery + Shenzhen (~800 images).
- Dental panoramic remains the most underserved modality.** Only two production-ready model families (`SerdarHelli` U-Net for segmentation, `SubGlitch1` YOLOv8n for pathology detection). Both have unclear or missing license headers.
- fastai pickle format in the DR classifier.** `jdelgado2002/diabetic_retinopathy_detection` ships as a fastai `.pkl`, which is a pickled Python object - it executes arbitrary code on load. Do not deploy it as-is on edge devices. Either re-export as TorchScript / ONNX inside a sandboxed pipeline, or retrain EfficientNet-B0 from the `mikelkl` notebook.
- ATOMMIC dependency for the brain segmentation model.** `wdika/SEG_UNet_BraTS2023AdultGlioma` uses the `.atomic` format, which requires the ATOMIC library to load. Convert to TorchScript at packaging time so deployments don't carry the dependency.
- License gaps are common in this ecosystem.** Half the lung, dental, vessel, and DR models above have no license file. Treat them as "research use only" until clarified. See the license matrix below.
- 3D networks are out of scope.** MONAI SegResNet, BraTS 3D U-Net, nnU-Net 3D pipelines exceed 8 GB peak memory on Pi 5 and run at 30+ s per volume. Always use the 2D-slice variant.
- ViT-Base and SigLIP-base are borderline.** Listed for completeness but practical latency on Pi 5 is 2-3 s even after INT8 quantization. Prefer CNNs (MobileNet, EfficientNet-B0, DenseNet121) wherever possible.

License Risk Matrix

License class	Models	Notes
Permissive (MIT / Apache-2.0)	alanjafari/BrainTumorAI, ayushirathour pneumonia, wdika brain seg, prithivMLmods bone, lizardwine derm, jdelgado2002 DR, Ahmed-Selem Shifaa	Safe for commercial use including closed-source distribution
AGPL-3.0 (network-share/copyleft)	RuiyangJu YOLOv8 fracture, keremberke YOLOv8m chest, all Ultralytics YOLOv8 pretrained weights	If RadPi is ever a hosted service, AGPL forces source release. Train your own YOLOv8 weights on the same datasets and re-license to MIT under your training pipeline, or switch to MobileNetV3 (mo26-web) under a permissive license.
GPL-3.0 (copyleft)	mdciri YOLOv7 fracture	Same concern as AGPL for closed-source distribution
No license stated	ianpan chest, imlab-iiip lung, SerdarHelli dental, SubGlitch1 dental, AndreyGermanov caries, orobix retina, VesselSeg-Pytorch, seva100 optic	Treat as “research only” until you contact the author. Several have HF Space demos which implicitly suggest non-commercial intent.
Notebook / training code only	mo26-web bone, Mehrdad-Noori brain seg, MALUNet derm, mikelkl APTOS	License attaches to your retrained weights, not the original; you control distribution

Conclusion

The MobileNetV2 / EfficientNet-B0 / YOLOv8n architecture family consistently delivers the best accuracy-to-size ratio for Raspberry Pi 5 deployment across all six medical imaging domains. Dermatology and chest X-ray are saturated with mature pretrained options. Brain MRI, fundus, and bone fracture detection each have one strong primary pick that ships ready-to-run. Dental panoramic and bone fracture segmentation remain genuine gaps requiring either custom training or workaround strategies (Grad-CAM, binary instead of per-instance segmentation). For every module, INT8 post-training quantization to ONNX or TFLite is mandatory: it halves inference latency, quarters model size, and keeps the full six-modality pipeline well under the Pi 5's 8 GB RAM ceiling.